

Question Sheet (Pinak Mahapatra)

Question 1

Consider a linear regression problem in which you are given samples

$$\{(x_i, y_i) : 1 \leq i \leq n\}$$

to construct the training model. Suppose that you scale one of the predictors (features) by a factor of 2 before building an optimized model to find the weight vector w . Will the optimal value of the least-squares loss change because of this modification to the training data? Why or why not?

Question 2

For each of parts (a) through (d), indicate whether we would generally expect the performance of a **flexible** statistical learning method to be better or worse than an **inflexible** method. Justify your answer.

- (a) The sample size n is extremely large, and the number of predictors p is small.
- (b) The number of predictors p is extremely large, and the number of observations n is small.
- (c) The relationship between the predictors and response is highly non-linear.
- (d) The variance of the error terms, i.e., $\sigma^2 = \text{Var}(\varepsilon)$, is extremely high.

Question 3

What are the advantages and disadvantages of a very flexible (versus a less flexible) approach for regression or classification? Under what circumstances might a more flexible approach be preferred to a less flexible approach? When might a less flexible approach be preferred?

Hint: Inflexible is more interpretable, fewer observations required, can be biased. Flexible can overfit (high error variance). In cases where we have high n or non-linear patterns flexible will be preferred.

Question 4

Describe the differences between a parametric and a non-parametric statistical learning approach. What are the advantages of a parametric approach to regression or classification (as opposed to a non-parametric approach)? What are its disadvantages?

Question 5 (Programming Question)

Using Python (scikit-learn), generate a synthetic regression dataset with $n = 200$ observations and $p = 5$ predictors.

- (a) Fit a **linear regression model** (parametric, inflexible).
- (b) Fit a **decision tree regressor** (non-parametric, flexible).
- (c) Compare their test mean squared errors (MSE) on a held-out test set.
- (d) Plot the training error and test error of the decision tree model as a function of tree depth.
- (e) Using the plot, explain how the **bias–variance trade-off** is illustrated. At what depth do you see signs of overfitting?

- (f) Discuss under what conditions the flexible method outperforms the inflexible one, and vice versa.